From Physical Energy to Market Regimes: Cross-Domain PI-MoE for Financial State Extraction

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Abstract

Reliable market-state identification requires representations that remain informative even when return innovations are close to a random walk and forecasting losses are dominated by conservative baselines. We study PI-MoE, a physics-informed mixture-of-experts model pretrained on physical time series and transferred to equity panels, and evaluate transferred gate logits as regime representations. On the main 2×4 equity panel, the zero-shot physical-pretrained gate reaches 0.696 ± 0.0342 AUROC across five random seeds, and a 10% linear probe raises this value to 0.834 ± 0.0060, outperforming both target-only and random-initialized MoE gates in regime alignment. The panel ablation shows that broader cross-sections of returns are not enough: regime separation is weakest on return-only panels, improves when liquidity is added, and is strongest when trade activity, fund participation, and balance-sheet information are observed together. These results support a narrower claim than forecasting superiority. In the present setting, physical pretraining is most useful when it transfers a stress-sensitive state geometry that can be read with little target supervision.

Keywords: time-series foundation models; market regime extraction; mixture-of-experts; cross-domain transfer; financial state detection

Area: Data-Driven Analytics

1 Introduction

Financial applications often need a stable ranking of market stress before they need a marginal improvement in one-step return prediction. Risk limits, surveillance thresholds, and hedging intensity are typically adjusted when volatility and liquidity conditions change, even when next-period returns remain difficult to forecast. That distinction matters in cross-domain transfer because financial innovations are close enough to a random walk that squared-error losses systematically favor conservative baselines such as Persistence or ARIMA. A transferred model can therefore be useful without winning a forecasting contest if it places calm and stressed observations in a consistent state ordering.

The finance literature has long treated market states as heterogeneous. Regime-switching models document recurrent changes in volatility, persistence, and dependence [Hamilton, 1989, Ang and Timmer-

mann, 2012]. Later work extends state monitoring to conditional covariance, connectedness, systemic-stress composites, and capital-shortfall measures [Engle, 2002, Billio et al., 2012, Holló et al., 2012, Brownlees and Engle, 2017]. In these settings, the informative signal rarely resides in raw returns alone. It appears in the joint behavior of volatility, co-movement, liquidity pressure, and balance-sheet conditions. What remains less clear is whether such state structure must always be learned within finance or whether some of it can be inherited from another domain.

Recent machine-learning evidence makes that question more concrete. Deep models and ensemble methods can extract nonlinear structure in cross-sectional returns, trading signals, and high-frequency price formation [Krauss et al., 2017, Fischer and Krauss, 2018, Gu et al., 2020, Sirignano and Cont, 2019]. At the same time, their gains are uneven: improvements in short-horizon prediction do not automatically translate into stable monitoring signals, interpretable risk control, or broad economic value [Heaton et al., 2017, Engle et al., 2013, Giglio et al., 2022, Wang et al., 2025]. That mixed record is important here. It suggests that a transferred model should not be judged only by whether it reduces innovation MSE relative to conservative baselines.

Time-series transfer research provides a plausible route for transferring something narrower than a full forecast rule. RevIN addresses distribution shift at the series level [Kim et al., 2022]; sparse MoE routing allows different temporal patterns to activate specialized experts [Shazeer et al., 2017]; and decomposition-based models such as N-BEATS, Autoformer, FEDformer, PatchTST, and MOMENT show that reusable structure can persist across scales, tasks, and datasets [Oreshkin et al., 2020, Wu et al., 2021, Zhou et al., 2022, Nie et al., 2023, Goswami et al., 2024]. Most of that evidence, however, is still organized around forecasting benchmarks. Much less is known about whether a transferred routing manifold can be read directly as a financial state representation.

We study that question with **PI-MoE** (Physics-Informed Mixture-of-Experts), pretrained on ETTh1, ETTh2, and weather wind speed and transferred to equity panels constructed from returns, trade value, fund participation, and balance-sheet information. We use gate logits as transferred representations and test whether they separate realized-volatility states before and after light readout calibration. On the main 2×4 panel, the zero-shot physical-pretrained gate reaches 0.696 ± 0.0342 AUROC across five random seeds, and a 10% linear probe increases that value to 0.834 ± 0.0060. The ablation is consistent with two empirical expectations that motivate the study: return-only panels are too weak to anchor transferred state structure, whereas panels mixing fast, medium, and slow financial signals are more likely to yield readable regime geometry.

2 Related literature

The relevant literature is useful here not as a catalogue of adjacent methods but as a way to delimit what is already known and what remains unresolved. Three strands matter most for the analysis that follows: financial state identification and risk monitoring, machine learning in finance, and transferable time-series modeling.

2.1 Financial state identification and risk monitoring

Financial state identification begins from the observation that a single linear law does not describe the full range of market behavior. Regime-switching models formalize recurrent changes in volatility, persistence, and dependence [Hamilton, 1989, Ang and Timmermann, 2012], while hidden-state methods such as HMMs and Gaussian mixtures remain useful because they can recover latent partitions without explicit labels [Rabiner, 1989, Bishop, 2006]. What these models establish is not only that market states exist, but also that their interpretation depends on conditional variability, persistence, and cross-series dependence rather than on returns alone.

Later work pushes the same idea beyond latent returns. Dynamic conditional covariance models, connectedness measures, systemic-stress composites, and capital-shortfall metrics all treat market conditions as multi-indicator objects [Engle, 2002, Billio et al., 2012, Holló et al., 2012, Brownlees and Engle, 2017]. In other words, useful state monitoring is increasingly built from co-movement, stress transmission, and risk concentration rather than from a single price process. That evidence matters here because it implies that a transferable financial state representation should be judged by its alignment with stress diagnostics. At the same time, these studies are overwhelmingly within-domain: they estimate states from financial inputs and then interpret them inside finance. They do not ask whether a representation learned outside finance can survive transfer as a usable financial state geometry.

2.2 Machine learning in finance: gains and limits

Machine-learning applications in finance have produced convincing local gains. Nonlinear models improve some cross-sectional return forecasts, some trading rules, and some high-frequency tasks [Krauss et al., 2017, Fischer and Krauss, 2018, Gu et al., 2020, Sirignano and Cont, 2019]. Together these studies show that financial data contain reusable nonlinear structure. They also suggest that richer models help most when the target is sufficiently local, the signal-to-noise ratio is not too low, or the evaluation target goes beyond raw point prediction.

The same literature also makes clear that these gains come with boundaries. Financial deep learning is often sensitive to horizon choice, model instability, and evaluation protocol [Heaton et al., 2017, Giglio et al., 2022]; macro and balance-sheet information can improve volatility and risk forecasts without eliminating instability [Engle et al., 2013]; and in risk-management settings, the most valuable models are often those that improve monitoring or tail-risk control rather than those that dominate on average predictive loss [Wang et al., 2025]. This mixed evidence matters for the way the study is framed. We do not treat universal forecasting gains as the only admissible sign of transfer. We ask instead whether physical pretraining can deliver a state representation that remains useful when the target forecasting problem itself is close to a martingale benchmark.

2.3 Transferable time-series modeling

Transfer-oriented time-series modeling focuses on several reusable mechanisms rather than on a single architecture family. RevIN addresses instance-level distribution shift [Kim et al., 2022], foundation-style pretraining demonstrates that large temporal representations can be reused across datasets and tasks

[Goswami et al., 2024], and sparse MoE routing provides a way to specialize computation by temporal pattern rather than by a single global rule [Shazeer et al., 2017]. Together, these ideas make it plausible that what transfers across domains is not a full task solution but a reusable internal organization of temporal structure.

Decomposition-based models strengthen that interpretation. Wavelet methods separate signals by scale [Mallat, 1999]; N-BEATS turns basis expansion into an interpretable forecasting backbone [Oreshkin et al., 2020]; and transformer models such as Autoformer, FEDformer, and PatchTST show that decomposition, frequency structure, and token-level patching can improve long-range temporal reuse [Wu et al., 2021, Zhou et al., 2022, Nie et al., 2023]. Yet most of this evidence is still benchmarked through forecasting quality or generic transfer performance. The unresolved question here is narrower and more finance-specific: once a routing structure is learned on physical data, can it be transferred into finance and read with little supervision as a representation of market states rather than merely as a forecasting aid?

3 PI-MoE and transferred gate readout

The PI-MoE architecture is described here only to the extent needed for the transfer argument. The emphasis is on the parts of the model that determine whether a routing structure learned on physical systems can remain readable after transfer to finance.

3.1 Problem formulation and architecture overview

Let $x_{t-L+1:t} \in \mathbb{R}^{L\times C}$ denote a multivariate context window of length L with channel count C, where the channel configuration is determined by the chosen panel design. In the physical source domains, PI-MoE is trained as a multi-scale innovation forecaster. After transfer to the financial target domain, we do not judge the model primarily by forecast loss. Instead, we test whether its internal routing representation induces a coherent ordering of financial stress. Regime labels are therefore defined from realized-volatility-based partitions, and the empirical question becomes whether the transferred representation separates high- and low-volatility states in a stable way.

PI-MoE combines instance-level normalization, wavelet decomposition, sparse routing, and diffusion-based expert modeling in a single architecture. Given a context window $x_{1:L}$, the model first applies RevIN in source-domain training, then decomposes the signal into db4 wavelet bands, routes the resulting multi-frequency representation through a sparse mixture of diffusion experts, and finally reconstructs a future innovation path. In the main financial experiments, target-domain RevIN is disabled because the target panel mixes variables with different economic memory speeds. The pretrained backbone is kept intact after transfer, and the empirical analysis centers on whether the gate retains a usable state geometry.

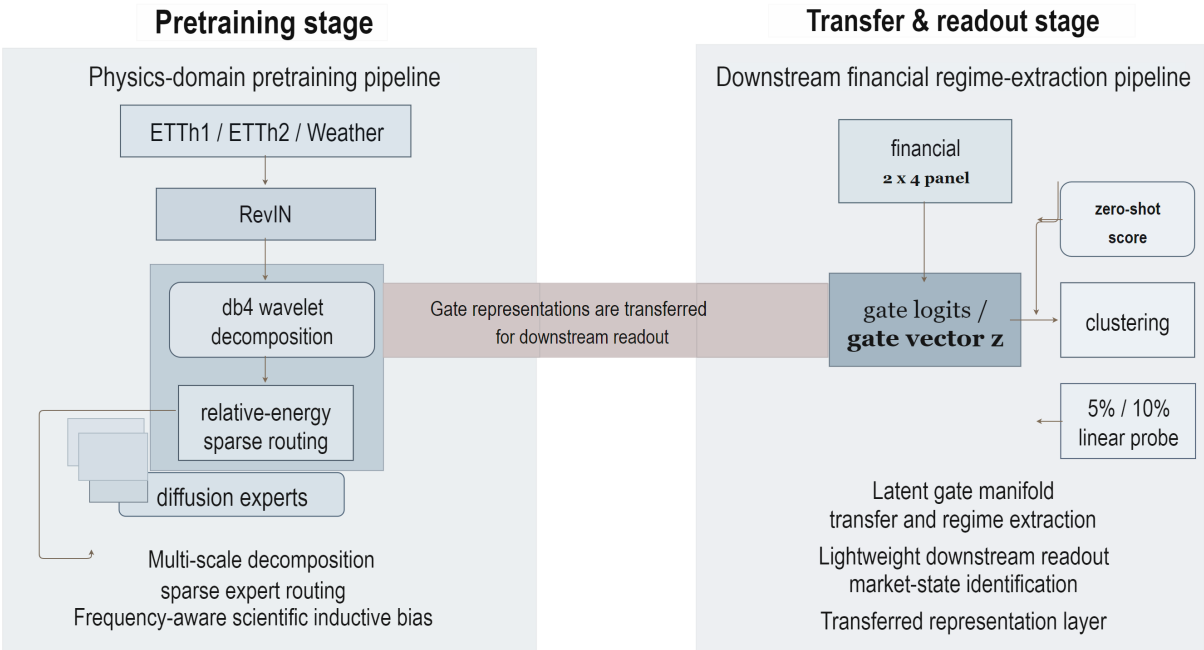

Figure 1: Cross-domain PI-MoE workflow. The left panel shows source-domain pretraining on physical time series with wavelet decomposition, sparse routing, and diffusion experts. The right panel shows transfer of gate logits to the 2×4 equity panel, followed by zero-shot scoring, gate-space clustering, and linear-probe readout.

3.2 Frequency-aware sparse routing

The routing mechanism is designed to depend on relative, rather than absolute, spectral content. After wavelet decomposition, the model computes band energy

$$E_j = \sum_t |c_j(t)|^2, \qquad r_j = \frac{E_j}{\sum_\ell E_\ell}. \tag{1}$$

The routing input is then constructed from the log-relative energy vector together with a spectral-entropy summary,

$$u_t = [\log(r_1 + \epsilon), \dots, \log(r_B + \epsilon), H(r)]. \tag{2}$$

The gate maps this descriptor into sparse top-2 weights

$$g_t = \mathrm{Top2}(\mathrm{MLP}(u_t)), \tag{3}$$

which are used to combine the experts' denoising outputs. This construction is important for transfer because it encourages the gate to respond to the balance among frequencies rather than to domain-specific scale levels. In the physical source domains, this balance reflects periodic and persistent structure; in the financial target domain, it becomes a candidate carrier of regime information.

3.3 Transferred representations and readout strategies

The original architecture was designed to forecast future innovations through expert aggregation and a light output adapter. Here the backbone is kept intact, but the empirical focus shifts from the forecast itself to the gate manifold. Let $\ell_{t,c,e}$ denote the pre-softmax gate logit for time index t, channel c, and expert e. We define a sample-level gate representation by averaging over channels:

$$z_t = \frac{1}{C} \sum_{c=1}^{C} \ell_{t,c,:}. \tag{4}$$

The vector z_t is passed to all downstream state-readout procedures. Instead of introducing a heavy target-side adapter, we ask whether transferred gate logits already contain a state geometry that can be exposed by light readouts. We therefore evaluate the same transferred representation with three readout strengths: zero-shot scoring, unsupervised gate-space partitioning, and few-shot linear probes. These readouts are compared with target-only and random-initialized gate baselines so that any gain can be attributed to physical pretraining rather than to the generic flexibility of an MoE architecture.

Zero-shot readout. The most direct test uses the transferred gate without any target-side end-to-end retraining. A zero-shot regime score is derived from the expert margin associated with the volatility-aligned expert pair. This construction asks whether physical pretraining alone already orders financial observations by stress intensity.

Gate-space clustering. To examine whether the transferred manifold is intrinsically clusterable, we apply k-means and Gaussian mixture models directly to z_t. These procedures do not alter PI-MoE itself; rather, they test whether unsupervised algorithms can recover high- and low-volatility partitions from the transferred geometry alone.

Few-shot linear probes. The main few-shot adaptation is intentionally light. Given support samples selected from low- and high-volatility extremes, we fit a linear probe

$$s_t = w^\top z_t + b, \tag{5}$$

using either 5% or 10% of the available support set. The rationale is straightforward. If the transferred manifold is already meaningful, then a simple linear separator should be sufficient to sharpen it. If, by contrast, useful structure emerges only after heavy adaptation, then the main source of performance would no longer be the transferred gate itself.

Direct gate baselines. Finally, the physical-pretrained gate is compared with two direct baselines. The first is a target-only PI-MoE gate trained on the financial panel without physical pretraining. The second is a random-initialized MoE gate. These baselines clarify whether any observed advantage comes from physical pretraining itself rather than from the generic flexibility of a sparse MoE architecture.

Table 1: Source and target configurations used in the equity transfer study.

Setting	Ch.	Financial feature set	Role in the paper
Source pretraining	1	ETTh1/ETTh2 OT; weather wind speed	Pretraining data for the frequency-conditioned experts.
2x4 main panel	8	2 stocks x return, trade value, fund count, assets	Main panel for the multi-seed regime results.
4x2 return+trade	8	4 stocks x return, trade value	Ablation on return and liquidity signals.
4x2 return+assets	8	4 stocks x return, assets	Ablation on return and slow balance-sheet signals.
8x1 return-only	8	8 stocks x return	Ablation with broader coverage and minimal feature diversity.

Notes: This table reports the source and target configurations used in the equity transfer study. All target configurations keep the channel budget fixed at eight while varying feature richness and cross-sectional breadth. The 2x4 panel is the main specification used for multi-seed regime results, whereas the remaining panels are used for ablation analysis.

4 Experimental design

4.1 Data and panel configurations

The source domains are ETTh1, ETTh2, and weather wind speed, chosen to expose the model to physical systems with persistent low-frequency structure. The target domain is a family of daily equity panels assembled from stock returns, trade activity, fund participation, and balance-sheet features.

The main target panel is the 2×4 design. It keeps the eight-channel budget fixed while allocating capacity to two stocks and four heterogeneous features: return, trade value, fund count, and assets. This is the panel on which all formal claims are made. The remaining three panels are used only to identify which ingredients are necessary for regime emergence.

4.2 Evaluation framework

Benchmark families. The empirical comparison is organized into four benchmark families. The first is the *physical-pretrained zero-shot gate*, which scores the target domain directly from the transferred gate and therefore provides the cleanest test of cross-domain regime emergence. The second applies *gate-space clustering*, namely k-means and GMM, to the transferred manifold in order to test whether its structure is intrinsically clusterable even without supervised readout tuning. The third contains the *direct gate baselines*, namely a target-only PI-MoE gate and a random-initialized MoE gate, which separate the contribution of physical pretraining from the generic flexibility of an MoE architecture. The fourth is *few-shot calibration*, implemented through linear probes trained on 5% or 10% support samples. Forecasting metrics are reported only as supplementary diagnostics against Persistence, ARIMA, and target-only PI-MoE and are not used as the primary acceptance criterion of the study.

Metrics and reporting. The main regime-detection metrics are AUROC and F1 for high-volatility state recognition, together with Spearman correlations against realized volatility (RV), bipower variation (BV), jump variation, and jump intensity. This metric suite assesses whether a state representation produces a coherent ordering of market stress rather than merely a convenient classification score. All formal claims on the main panel are reported from five random seeds. Single-run experiments are used only to illustrate

Table 2: Main regime-detection results on the 2x4 panel across five random seeds.

Method	AUROC	F1	RV rho	BV rho	JumpVar rho	JumpInt rho
Physical-Pretrained Gate	0.696 ± 0.0342	0.746 ± 0.0243	0.246 ± 0.0669	0.247 ± 0.0798	0.342 ± 0.0646	−0.386 ± 0.0693
Physical-Pretrained + Gate-space k-means	0.745 ± 0.112	0.652 ± 0.0686	0.299 ± 0.158	0.267 ± 0.178	0.415 ± 0.188	−0.430 ± 0.216
Physical-Pretrained + Gate-space GMM	0.762 ± 0.120	0.664 ± 0.148	0.315 ± 0.165	0.292 ± 0.180	0.433 ± 0.200	−0.440 ± 0.220
Random-init MoE Gate	0.587 ± 0.150	0.744 ± 0.0412	0.111 ± 0.230	0.0940 ± 0.237	0.187 ± 0.231	−0.183 ± 0.211
Target-only MoE Gate	0.585 ± 0.0383	0.743 ± 0.0366	0.138 ± 0.0588	0.153 ± 0.0688	0.163 ± 0.0662	−0.172 ± 0.0764
Physical-Pretrained + Linear Probe (5%)	0.831 ± 0.0146	0.621 ± 0.0602	0.419 ± 0.0267	0.407 ± 0.0234	0.532 ± 0.0247	−0.588 ± 0.0104
Physical-Pretrained + Linear Probe (10%)	0.834 ± 0.0060	0.649 ± 0.0145	0.427 ± 0.0099	0.410 ± 0.0083	0.544 ± 0.0076	−0.591 ± 0.0106

Notes: This table reports the main regime-detection results on the 2x4 panel. Entries are mean ± standard deviation over five random seeds. Higher AUROC, F1, and rank correlations indicate stronger alignment with market stress. RV denotes realized volatility, BV denotes bipower variation, JumpVar denotes jump variation, and JumpInt denotes jump intensity. Gate-space k-means and GMM are applied to gate representations rather than to raw financial inputs.

the mechanism behind the multi-seed averages and not to replace them. All experiments are implemented in PyTorch on the same PI-MoE codebase used for source-domain pretraining. The empirical analysis is organized around regime emergence on the main panel, the ablation against forecasting proximity, and the practical boundary of the transfer claim.

5 Results and analysis

We begin with the 2×4 panel because it is the only target design evaluated with five-seed regime results and because it is the strongest configuration in the panel ablation. We then use the remaining panels and the forecasting supplement to identify where the transfer result strengthens, where it weakens, and what kind of downstream use it supports.

5.1 Regime emergence on the main panel

On the 2×4 panel, the physical-pretrained gate reaches 0.696 ± 0.0342 AUROC across five random seeds. The corresponding values for the random-initialized and target-only MoE gates are 0.587 ± 0.150 and 0.585 ± 0.0383. A 5% linear probe raises the mean AUROC to 0.831 ± 0.0146, and a 10% probe increases it further to 0.834 ± 0.0060 while keeping seed dispersion small.

Table 2 and figure 2 show that the AUROC gain is accompanied by stronger alignment on the stress diagnostics. Relative to the two direct gate baselines, the physical-pretrained gate has higher mean correlations with realized volatility, bipower variation, and jump variation, and the 10% probe improves those correlations again while lifting AUROC. The result is therefore not limited to a single thresholded classifier metric.

Figure 3 shows the same contrast in the representative run. The zero-shot physical-pretrained gate already forms a visible high- versus low-volatility gradient, whereas the random-init and target-only gates remain more diffuse. The 10% probe aligns a linear boundary with that existing geometry rather than introducing a completely new partition. The single-run values, 0.740 for zero-shot and 0.834 after the 10% probe, are reported only as a visual counterpart to the multi-seed evidence; the formal claim remains the five-seed average.

The readout hierarchy clarifies how much target-side calibration is actually needed once the manifold has been transferred. Gate-space GMM reaches 0.762 ± 0.120, which is higher than the zero-shot gate but

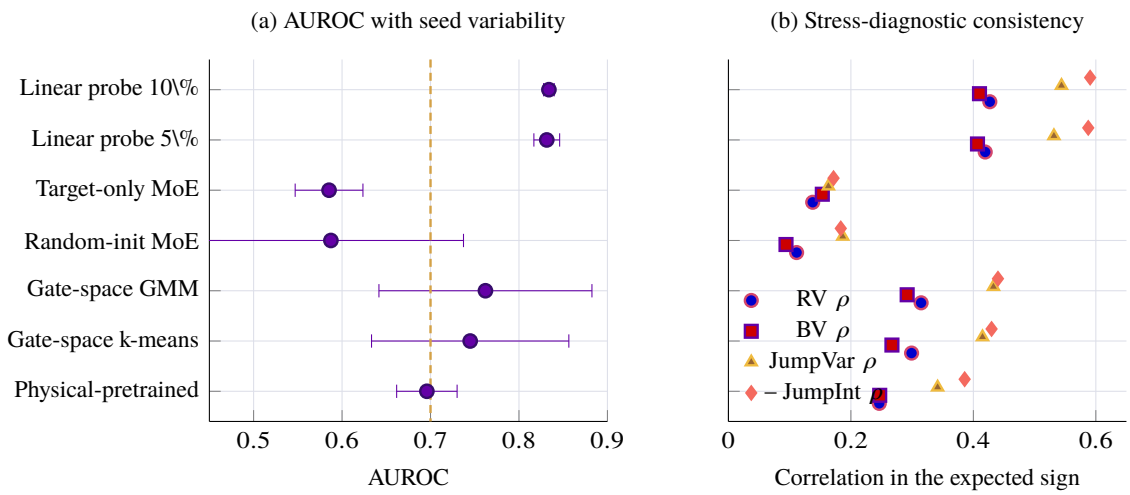

Figure 2: Main regime-detection results on the 2×4 panel. Panel (a) reports AUROC with seed-level standard-deviation error bars. Panel (b) reports the four stress-sensitive rank correlations used in the paper; jump-intensity is sign-flipped so that values farther to the right consistently indicate stronger alignment with stress.

still below the few-shot linear probes. The 5% and 10% probes reach 0.831 ± 0.0146 and 0.834 ± 0.0060, respectively, while the unsupervised clusterers show visibly larger seed dispersion in Table 2.

The ordering is informative because the same gate representation is being read in different ways. Zero-shot transfer establishes that structure is present. Gate-space clustering shows that the geometry is clusterable, but its variance is materially larger than that of the linear probes. The few-shot probes use little target supervision, yet they deliver both higher mean AUROC and lower variability than unsupervised clustering. Figure 4 shows the same pattern more compactly: the physical-pretrained gate starts from a stronger zero-shot position and converts a small support set into a stable gain, whereas the random-init and target-only gates remain reference-level alternatives.

This comparison narrows the role of few-shot calibration. The target domain does not need a large adapter to manufacture useful structure from scratch. Most of the usable ordering is already present in the transferred manifold; the probe mainly aligns that manifold with the target volatility labels. Taken together, these results indicate that physical pretraining transfers a readable ordering of financial stress even before target-side end-to-end retraining, and that light supervision mainly stabilizes the readout of an already structured manifold.

5.2 Feature richness vs. forecasting proximity

The panel ablation isolates what information the target panel exposes while keeping the channel budget fixed at eight. The resulting differences are therefore attributable to feature composition and cross-sectional allocation rather than to a larger target representation.

The ranking in Figure 5 and table 3 is consistent across the main regime metrics. The 2×4 panel has the highest zero-shot AUROC and the highest 10% linear-probe AUROC. The 8×1 return-only panel is weakest, so additional stocks do not compensate for missing feature diversity. The 4×2 return+trade panel improves on the return-only setting, which indicates that liquidity contributes useful high-frequency

Table 3: Ablation results across target-panel designs.

Panel design	Zero-shot AUROC	Linear probe 10% AUROC	PI-MoE 10% innovation MSE	Gap vs Persistence	Gap vs ARIMA
2x4 return+trade+fund+assets	0.740	0.834	0.0806	+0.0022	+0.0022
4x2 return+trade	0.607	0.567	0.114	+0.0099	+0.0101
4x2 return+assets	0.532	0.518	5.73×10^{-4}	$+2.54 \times 10^{-6}$	$+7.63 \times 10^{-6}$
8x1 return-only	0.460	0.551	0.0011	$+1.14 \times 10^{-4}$	$+1.39 \times 10^{-4}$

Notes: This table reports the ablation results across target-panel designs. Zero-shot AUROC and linear-probe AUROC are taken from the corresponding single-run regime-evidence files, and innovation MSE is reported at horizon 24. Positive gaps indicate that the PI-MoE few-shot 10% model remains above the corresponding Persistence or ARIMA benchmark. Bold values identify either the strongest regime result or the smallest forecasting gap.

Table 4: Forecasting supplement across target-panel designs.

Method	2x4 main	4x2 return+trade	4x2 return+assets	8x1 return-only
Persistence	0.0783	0.104	5.70×10^{-4}	9.99×10^{-4}
ARIMA(p,1,0)	0.0784	0.104	5.65×10^{-4}	9.73×10^{-4}
Target-only upper	0.0803	0.108	5.80×10^{-4}	0.0010
PI-MoE zero-shot	0.439	0.459	0.362	0.587
PI-MoE few-shot 5%	0.0855	0.109	5.83×10^{-4}	0.0011
PI-MoE few-shot 10%	0.0806	0.114	5.73×10^{-4}	0.0011

Notes: This table reports the forecasting supplement across target-panel designs. All entries are innovation MSE values at horizon 24, where lower values indicate better predictive accuracy. The table is included to contextualize the regime evidence and is not used as the primary criterion for judging cross-domain transfer.

structure, but its regime scores remain well below those of the 2×4 panel. The 4×2 return+assets panel behaves differently again: it is not strong at regime extraction, yet it is the closest panel to Persistence and ARIMA in forecasting space.

The forecasting supplement locates the boundary of that regime result. On the main 2×4 panel, the few-shot 10% model reaches 0.0806, while Persistence and ARIMA remain lower at 0.0783 and 0.0784. The result therefore does not support forecasting superiority on the main panel.

The 4×2 return+assets panel sharpens this distinction. Its few-shot 10% innovation MSE is 5.73×10^{-4}, which is close to the classical baselines, but its regime scores remain modest at 0.532 under zero-shot transfer and 0.518 after a 10% probe. Slow-moving fundamentals seem to act as anchors in prediction space: they reduce large deviations and therefore shrink the MSE gap. That same anchoring does not automatically create a well-separated regime manifold. Forecast proximity and regime extraction are therefore related but not interchangeable outcomes.

The remaining panels provide the negative evidence. The 4×2 return+trade panel reaches 0.607 under zero-shot transfer and 0.567 after a 10% probe. The 8×1 return-only panel is lower still, at 0.460 and 0.551. Liquidity alone is not enough, and broader cross-sectional coverage without richer features is not enough either. Regime geometry is strongest only when trade activity, institutional participation, and balance-sheet information are observed together.

Taken together, the ablation identifies where the transferred gate can and cannot lock onto the target domain. Return-only panels do not expose enough persistent multi-band structure. Liquidity helps, but by itself it is not enough. Slow fundamentals narrow the forecasting gap, yet they do not produce a sharply separated regime manifold on their own. The strongest regime signal appears only when fast, medium, and slow financial variables coexist in the same panel. We use *System Energy* as shorthand for the first

pattern and *Resonance* as shorthand for the second. These labels summarize the ablation; they are not introduced as independent theoretical proofs.

5.3 Practical implications and boundary conditions

The practical implication is narrower than replacing classical forecasters. PI-MoE is better interpreted as a regime-sensitive representation layer for monitoring market-state changes. In surveillance, stress screening, or portfolio oversight, a user could read the zero-shot gate as a rough ranking of stress and then sharpen that ranking with a very small support set. Figure 6 shows one representative episode in which the stress diagnostics and the transferred gate rise in the same broad segment, while the calibrated score becomes easier to interpret operationally.

The evidence in this paper supports a bounded claim. PI-MoE does not outperform Persistence or ARIMA as a universal return forecaster. More stocks do not automatically improve transfer, and cleaner return-only panels are not better target domains for this problem. The evidence also comes from one family of daily equity panels with realized-volatility-derived labels, so it does not yet establish that the same gate geometry will align equally well with exogenous stress indices or with broader equity universes. The method is most useful when the target panel exposes persistent multi-frequency structure; when that structure is absent, the transferred gate becomes difficult to read and the regime advantage narrows quickly.

6 Conclusion

This paper asked whether a physics-pretrained time-series model can be transferred to finance as a market regime extractor. On the main 2×4 equity panel, the answer is yes in a specific and testable sense: the zero-shot physical-pretrained gate outperforms both target-only and random-initialized MoE gates, and a 10% linear probe produces the strongest and most stable regime-detection results across seeds. The result is not a forecasting win. It is evidence that a routing structure learned on physical dynamics can survive transfer as a stress-sensitive financial manifold.

The panel ablations show when that mechanism appears and when it fails. Return-only panels are too weak to support regime emergence. Slow fundamentals narrow the forecasting gap but do not by themselves create strong state separation. The strongest result appears when trade activity, fund participation, and balance-sheet information coexist, which is why the 2×4 panel serves as the empirical anchor of the paper and the other panels remain boundary conditions rather than competing main specifications.

For practice, the evidence supports using PI-MoE as a representation layer for market surveillance, stress screening, and other regime-sensitive tasks, not as a drop-in replacement for conservative return forecasters. For research, the next step is to test whether the same transferred gate aligns with exogenous stress proxies and whether the geometry remains useful in broader cross-sectional universes. Within the scope of the present experiments, the strongest conclusion is therefore narrower than forecasting superiority: physical pretraining contributes value in finance when it transfers state geometry that can be read with very little target supervision.

References

Andrew Ang and Allan Timmermann. Regime changes and financial markets. *Annual Review of Financial Economics*, 4:313–337, 2012. doi: 10.1146/annurev-financial-110311-101808.

Monica Billio, Mila Getmansky, Andrew W. Lo, and Loriana Pelizzon. Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3):535–559, 2012. doi: 10.1016/j.jfineco.2011.12.010.

Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

Christian Brownlees and Robert F. Engle. SRISK: A conditional capital shortfall measure of systemic risk. *The Review of Financial Studies*, 30(1):48–79, 2017. doi: 10.1093/rfs/hhw060.

Robert Engle. Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3):339–350, 2002.

Robert Engle, Eric Ghysels, and Bumjean Sohn. Stock market volatility and macroeconomic fundamentals. *The Review of Economics and Statistics*, 95(3):776–797, 2013.

Thomas Fischer and Christopher Krauss. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669, 2018.

Stefano Giglio, Bryan Kelly, and Dacheng Xiu. Factor models, machine learning, and asset pricing. *Annual Review of Financial Economics*, 14:337–368, 2022.

Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. MOMENT: A family of open time-series foundation models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 16115–16152, 2024. URL https://proceedings.mlr.press/v235/goswami24a.html.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020. doi: 10.1093/rfs/hhaa009.

James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.

J. B. Heaton, N. G. Polson, and J. H. Witte. Deep learning in finance. *Applied Stochastic Models in Business and Industry*, 33(1):3–12, 2017.

Dániel Holló, Manfred Kremer, and Marco Lo Duca. CISS – a composite indicator of systemic stress in the financial system. Working Paper Series 1426, European Central Bank, 2012. URL https://data.ecb.europa.eu/data/datasets/ciss/data-information.

Taesung Kim, Jinhee Kim, Yunwon Tae, Cheonbok Park, Jang-Ho Choi, and Jaegul Choo. Reversible instance normalization for accurate time-series forecasting against distribution shift. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=cGDAkQo1C0p.

Christopher Krauss, Xuan Anh Do, and Nicolas Huck. Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research*, 259 (2):689–702, 2017.

Stephane Mallat. *A Wavelet Tour of Signal Processing*. Academic Press, 1999.

Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=Jbdc0vTOcol.

Boris N. Oreshkin, Dmitri Carpov, Nicolas Chapados, and Yoshua Bengio. N-BEATS: Neural basis expansion analysis for interpretable time series forecasting. In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1905.10437.

Lawrence R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2017.

Justin Sirignano and Rama Cont. Universal features of price formation in financial markets: Perspectives from deep learning. *Quantitative Finance*, 19(9):1449–1459, 2019. doi: 10.1080/14697688.2019.1622295.

Shuai Wang, Qian Wang, Helen Lu, Dongxue Zhang, Qianyi Xing, and Jianzhou Wang. Learning about tail risk: Machine learning and combination with regularization in market risk management. *Omega*, 133:103249, 2025. doi: 10.1016/j.omega.2024.103249.

Haixu Wu, Jiehui Xu, Jianmin Wang, and Mingsheng Long. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. In *Advances in Neural Information Processing Systems*, volume 34, pages 22419–22430, 2021. URL https://papers.nips.cc/paper/2021/hash/bcc0d400288793e8bdcd7c19a8ac0c2b-Abstract.html.

Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, and Rong Jin. FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 27268–27286, 2022. URL https://proceedings.mlr.press/v162/zhou22g.html.

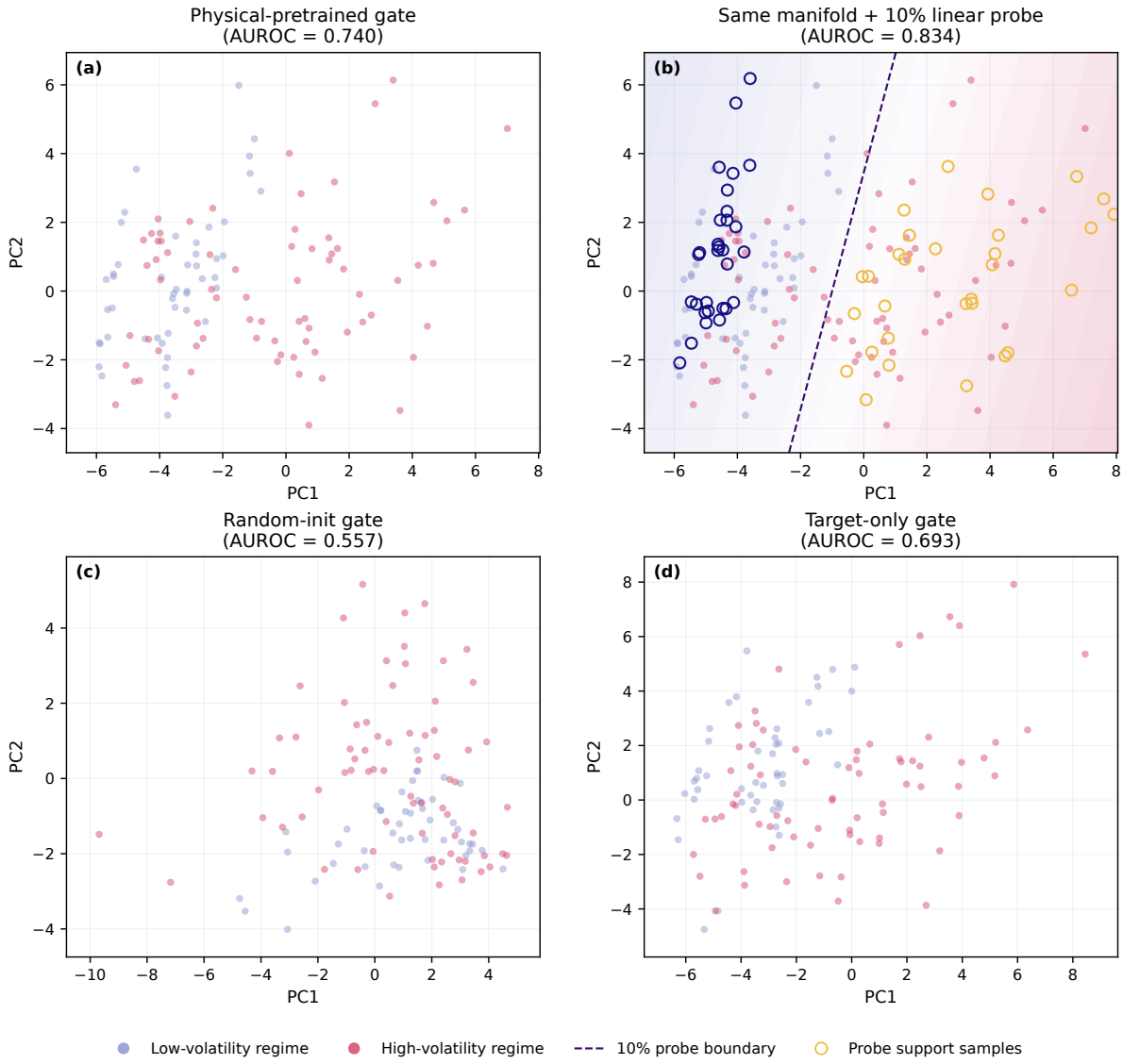

Figure 3: Representative gate-manifold geometry on the main 2×4 panel. Each panel shows a two-dimensional PCA projection of gate-manifold features from the representative single run used in the main experiment. The upper-left panel reports the zero-shot physical-pretrained gate, the upper-right panel overlays a 10% linear-probe decision boundary on the same projected manifold, and the lower panels report the corresponding manifolds from the random-initialized and target-only gates. The decision surface is shown only for visualization.

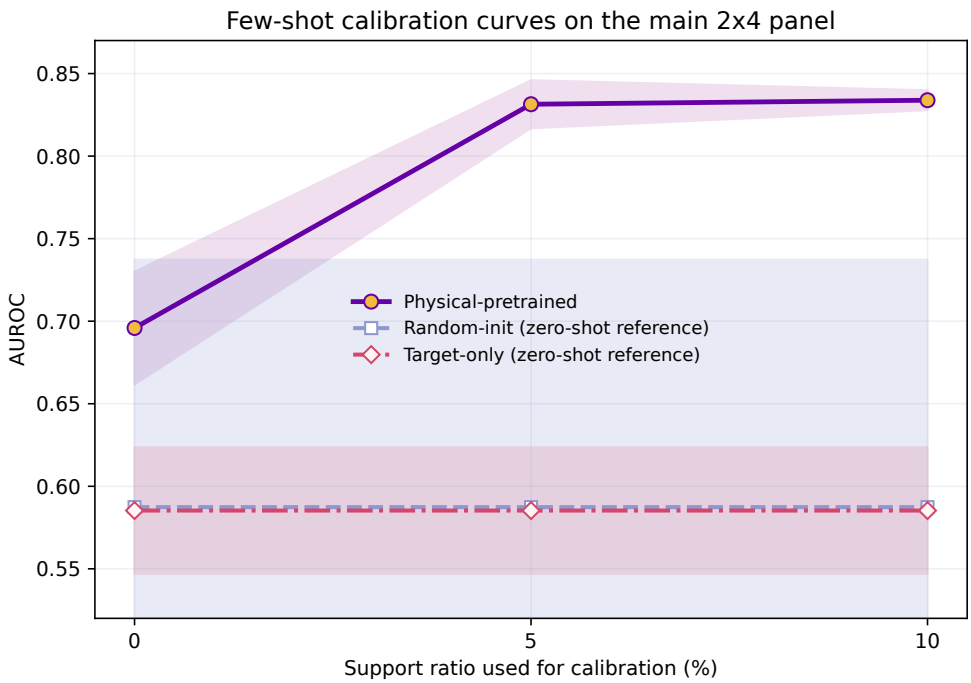

Figure 4: Few-shot calibration profile on the main 2×4 panel. The physical-pretrained curve reports the official five-seed mean AUROC at 0%, 5%, and 10% support ratios, together with one-standard-deviation bands. The random-init and target-only alternatives are shown as zero-shot reference bands because the formal few-shot protocol is centered on whether a transferred manifold can be sharpened by light calibration.

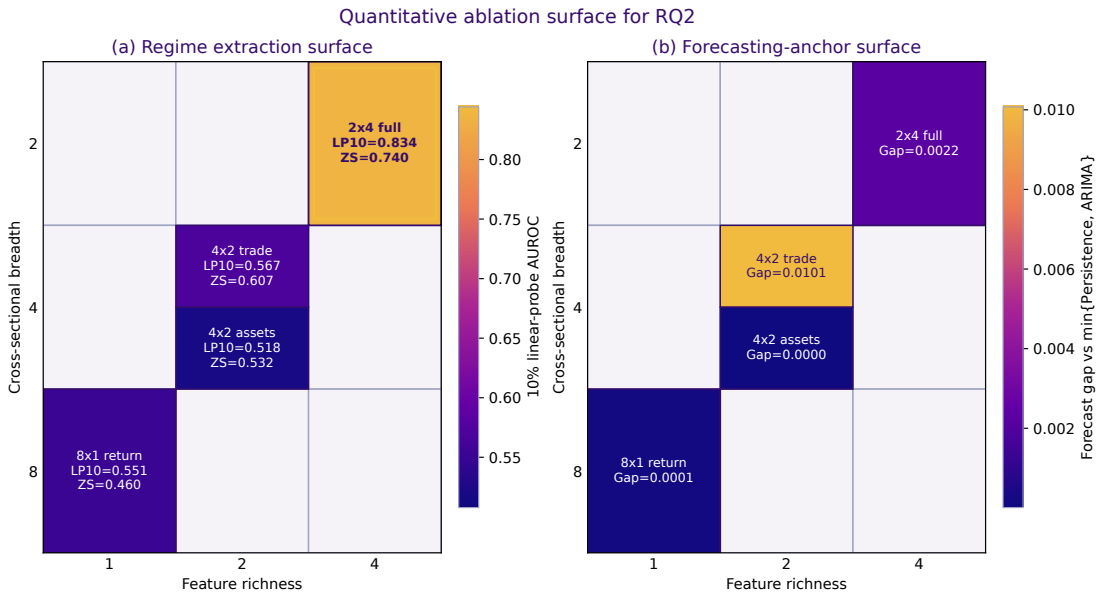

Figure 5: Quantitative ablation surface across panel designs. Panel (a) colors each observed panel design by its 10% linear-probe AUROC and annotates the corresponding zero-shot AUROC. Panel (b) colors the same designs by the innovation-MSE gap relative to the better of Persistence and ARIMA. The split cell at feature richness = 2 and breadth = 4 distinguishes the return+trade and return+assets panels.

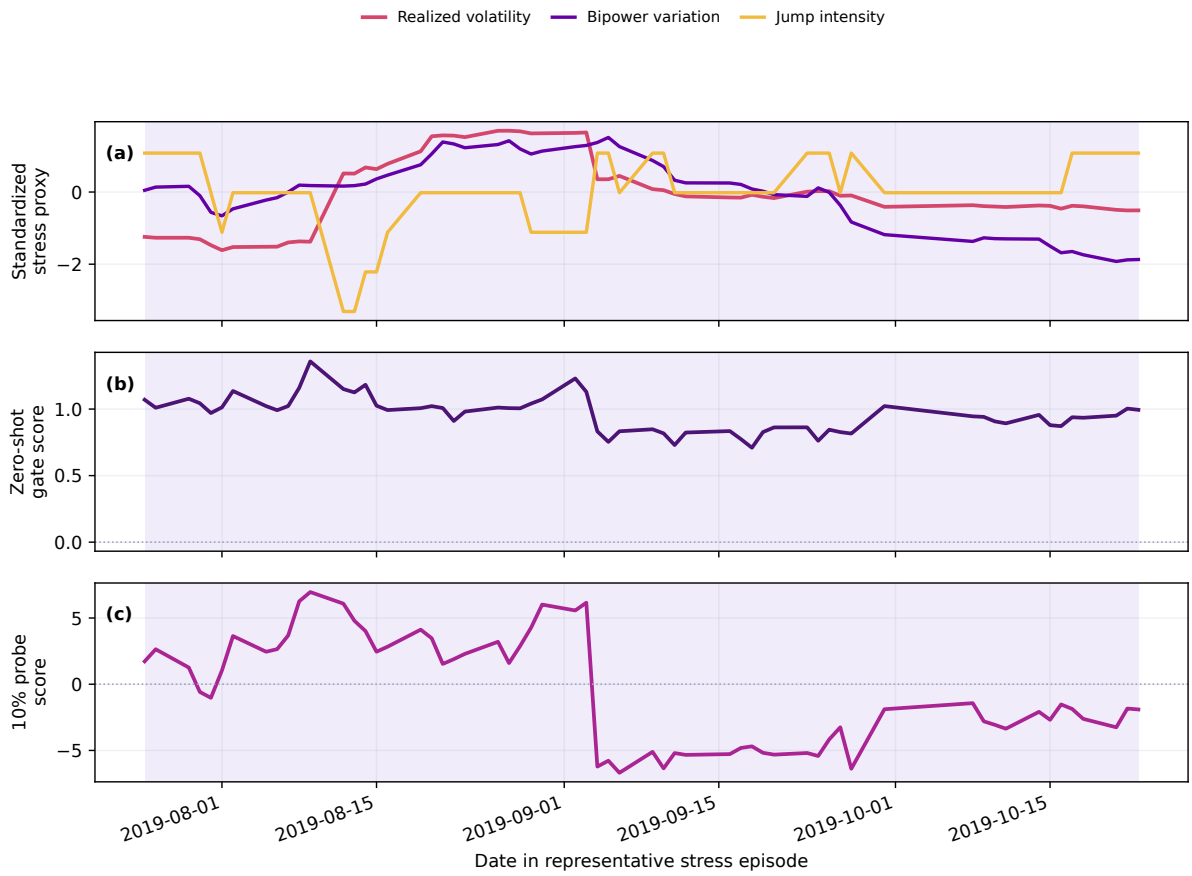

Figure 6: Representative stress episode on the main 2×4 panel. Panel (a) reports standardized realized volatility, bipower variation, and jump intensity; panels (b) and (c) report the corresponding zero-shot gate score and 10% linear-probe score. Gray bands indicate windows classified as high-volatility states by the realized-volatility threshold.